

Digital Communication System



Unit 2: Multiplexing Techniques and Network Switching

2.1 FDMA

2.2 TDMA

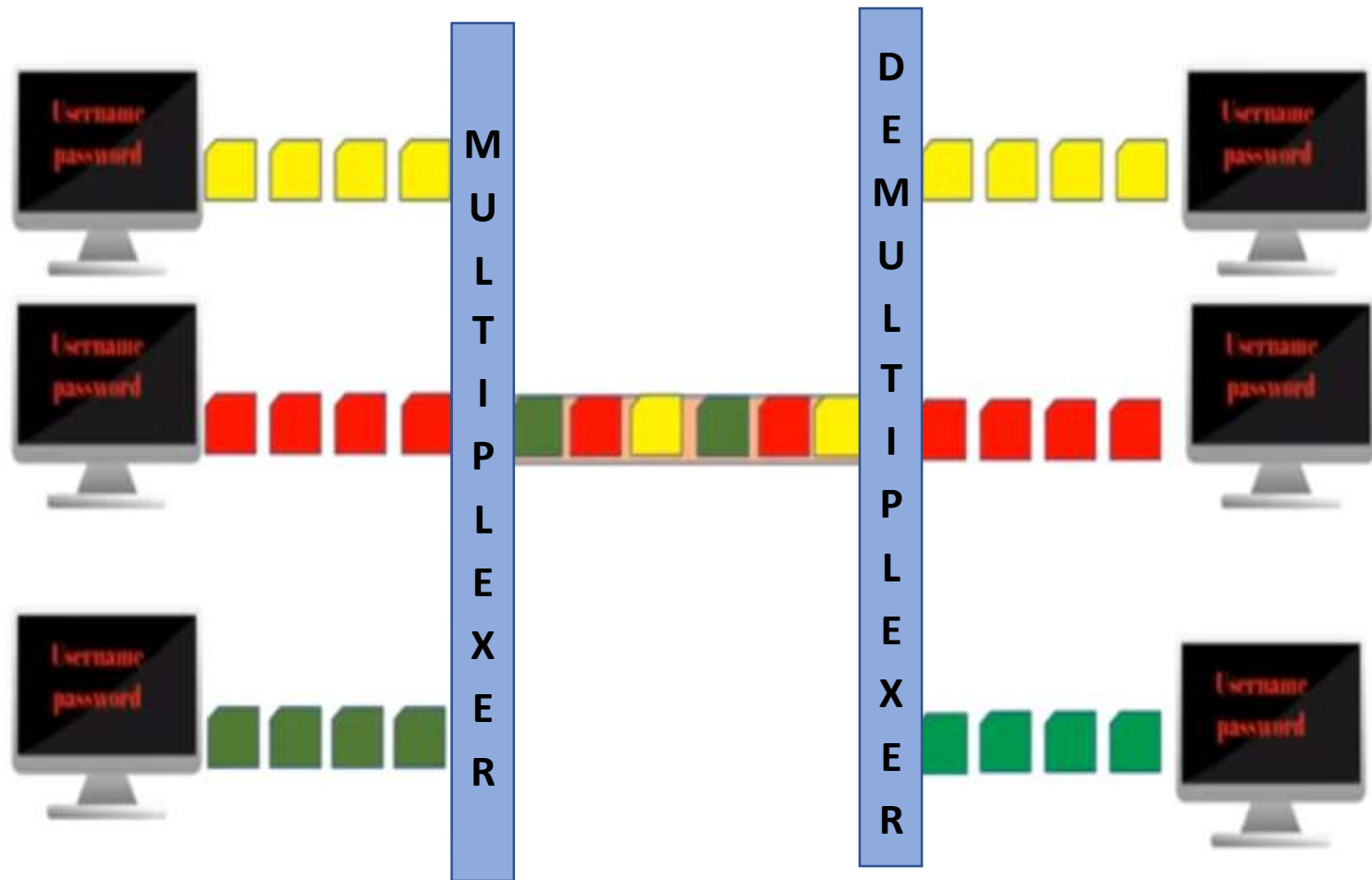
2.3 WDM

2.4 Circuit and Data (Packet) Mode, Circuit Switching, Packet Switching

2.5 Introduction of Transmission Media

What is Multiplexing?

Multiplexing is a set of techniques that allows simultaneous transmission of multiple signals on a single data link.



What is Multiplexing?

- **Multiplexing was first developed in telephony.** A number of signals were combined to send through a single cable. The process of multiplexing divides a communication channel into several number of logical channels, allotting each one for a different message signal or a data stream to be transferred. The device that does multiplexing can be called as **Multiplexer** or **MUX**.

What is Multiplexing?

The reverse process, i.e., extracting the number of channels from one, which is done at the receiver is called as **de-multiplexing**. The device that does de-multiplexing can be called as **de-multiplexer** or **DEMUX**.



Multiplexing and Demultiplexing

Types of Multiplexing

- If the analog signals are multiplexed, then it is called as **analog multiplexing**.

FDM – Frequency Division Multiplexing

WDM – Wavelength Division Multiplexing

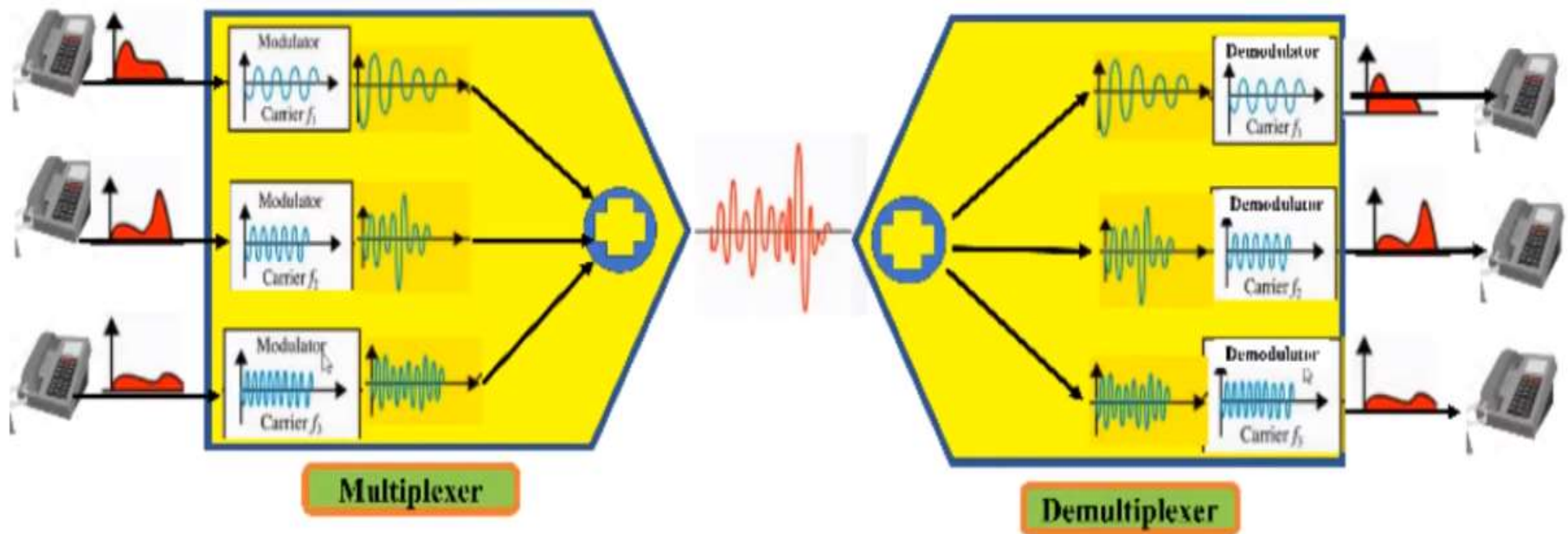
- If the digital signals are multiplexed, then it is called as **digital multiplexing**.

TDM – Time Division Multiplexing

FDM

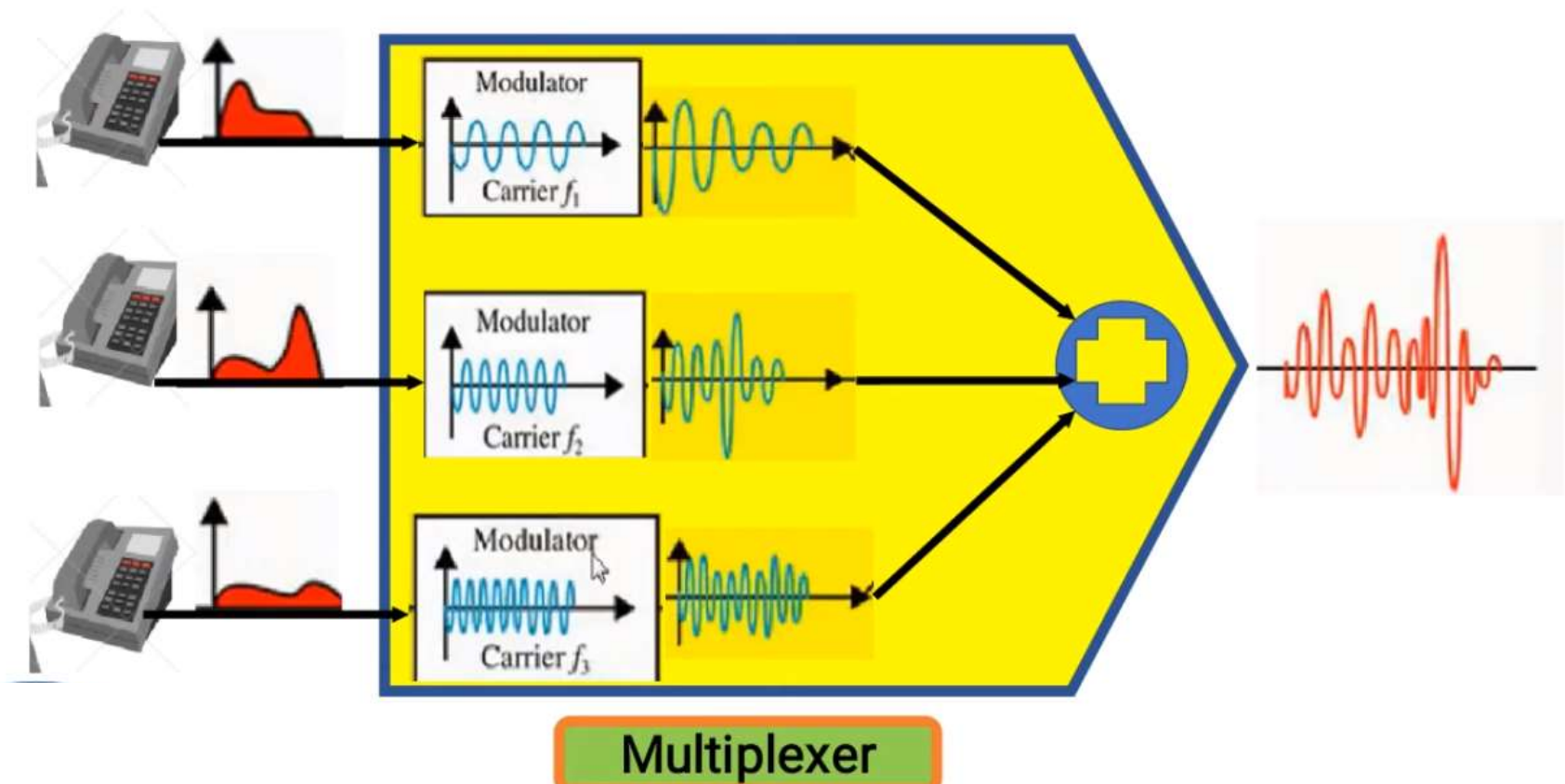
- **Frequency division multiplexing (FDM)** – It is a technique of multiplexing which means combining more than one **electronic signal over a shared medium**. In FDM, signals of different frequencies are combined for concurrent transmission.

FDM



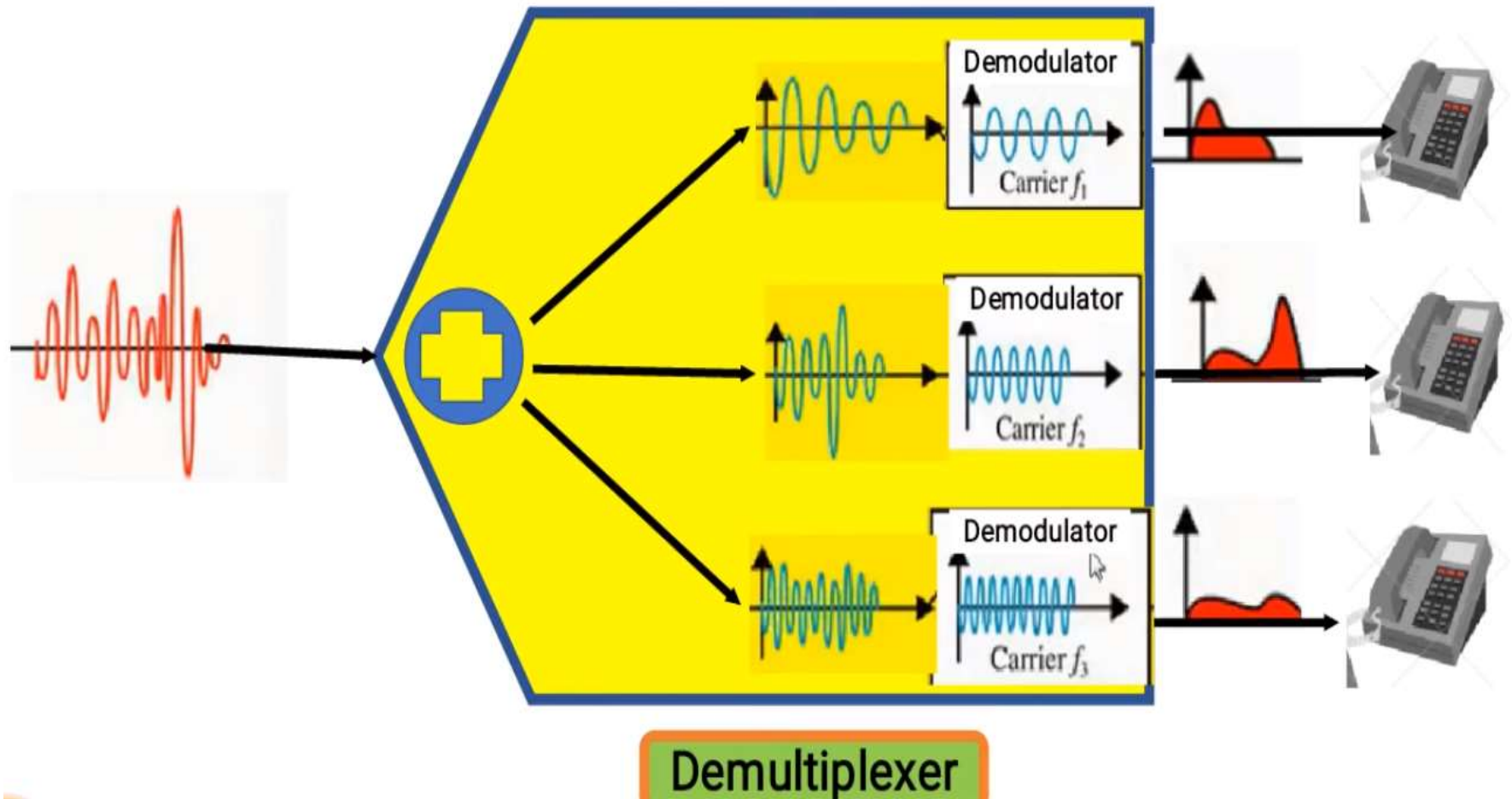
FDM

FDM: Multiplexing Process

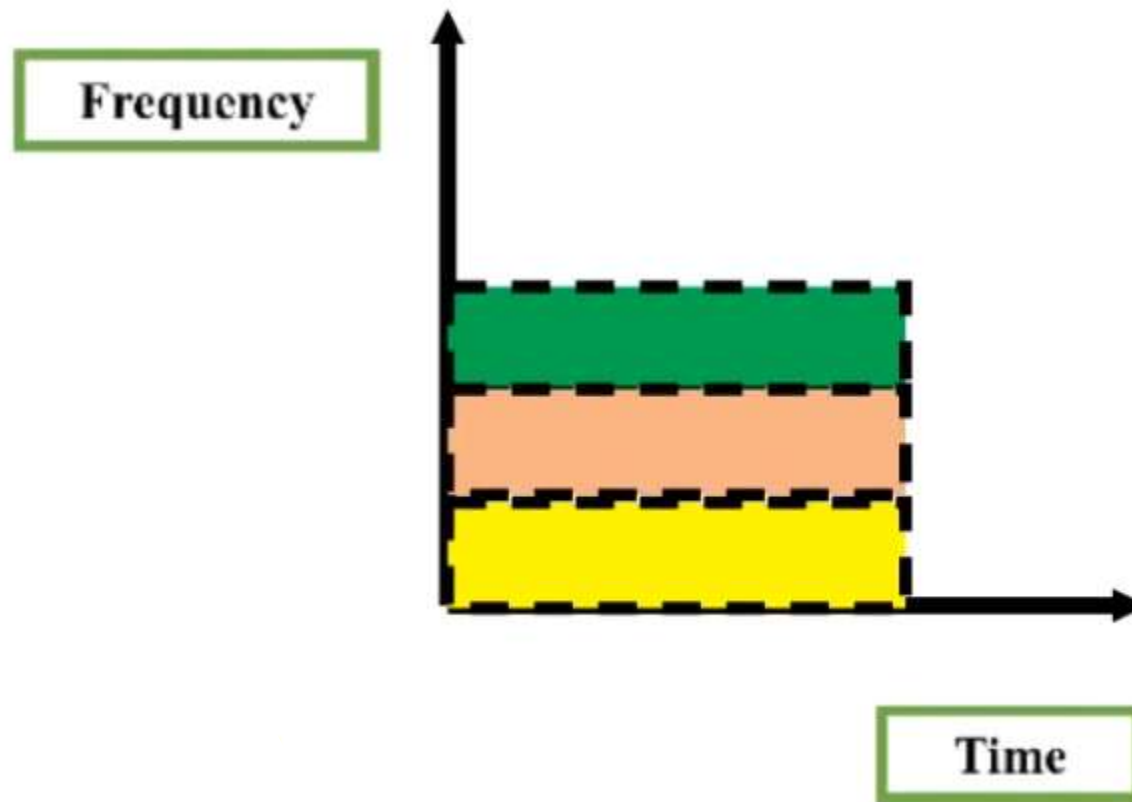


FDM

FDM: De-multiplexing Process



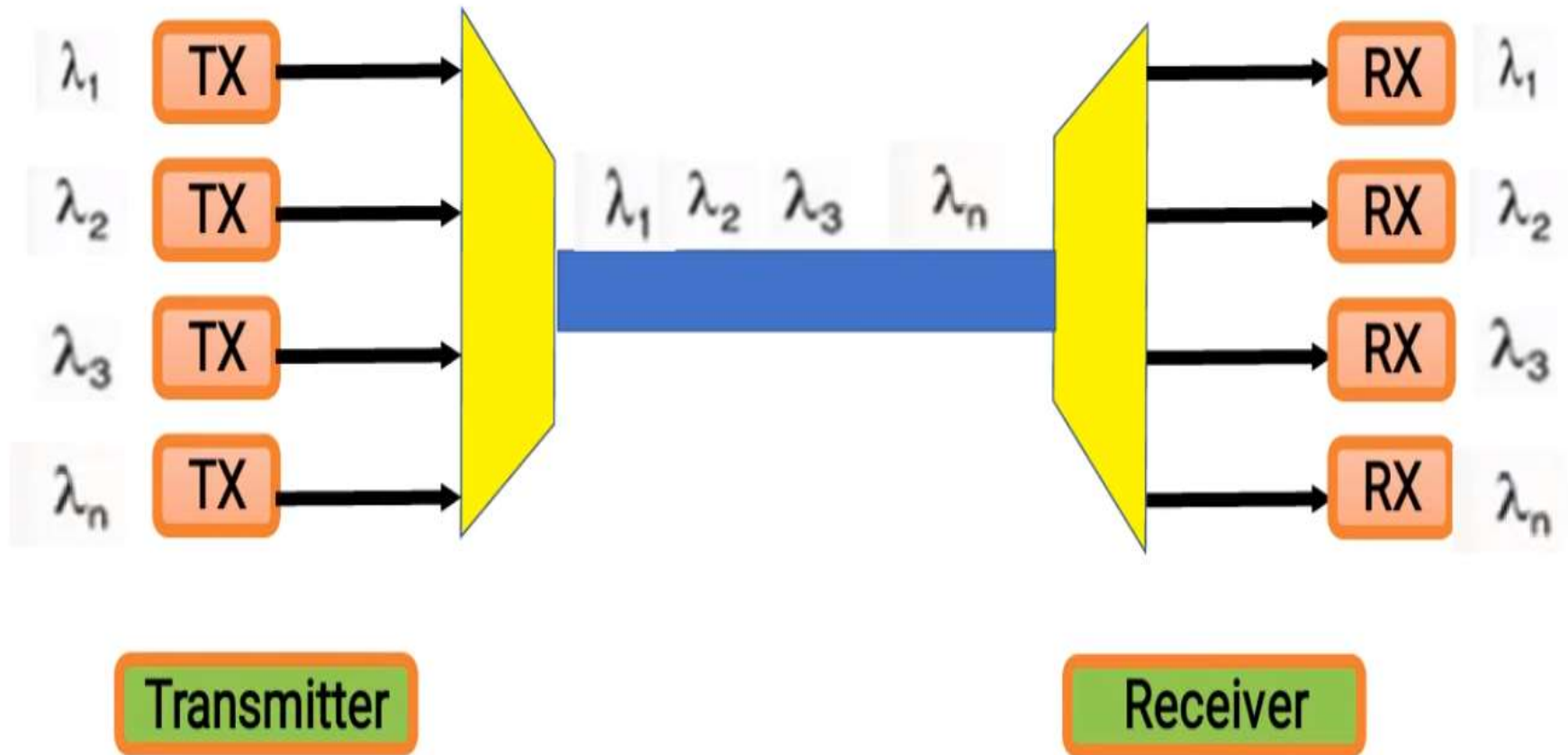
FDM



WDM

- **Wavelength Division Multiplexing** – Wavelength division multiplexing (WDM) is a technique of multiplexing **multiple optical carrier signals through a single optical fiber channel** by varying the wavelengths of laser lights. WDM allows communication in both the directions in the fiber optic cable.
- In WDM, the optical signals from different sources or (transponders) are combined by a multiplexer, which is essentially an optical combiner. They are combined so that their wavelengths are different.

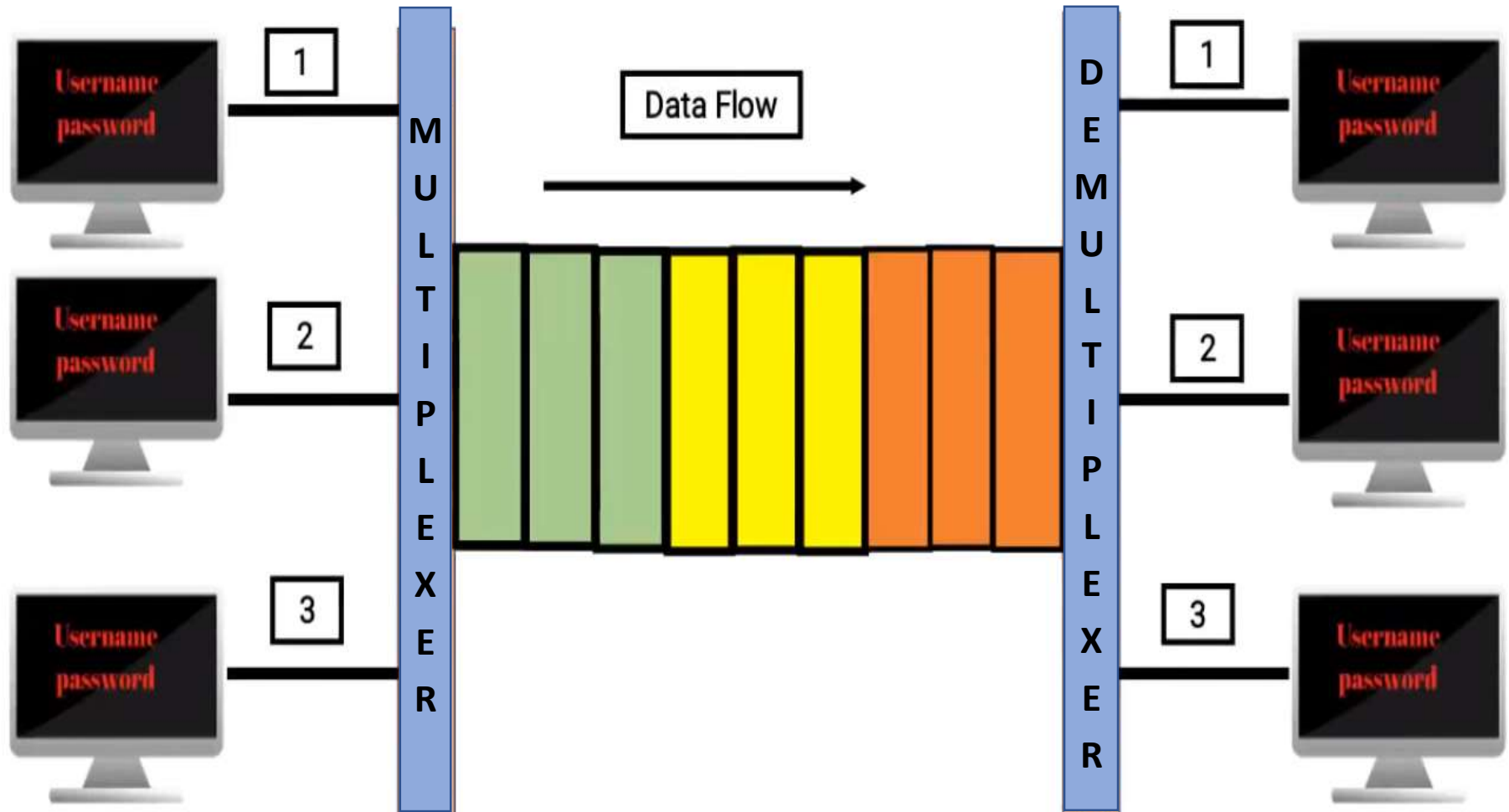
WDM



TDM

- **Time Division Multiplexing** - the time domain is divided into several recurrent slots of fixed length, and each signal is allotted a time slot on a **round-robin basis**.
- In TDM, the data flow of each input stream is divided into units. One unit may be 1 bit, 1 byte, or a block of few bytes. Each input unit is allotted an input time slot. One input unit corresponds to one output unit and is allotted an output time slot. During transmission, one unit of each of the input streams is allotted one-time slot, periodically, in a sequence, on a rotational basis. This system is popularly called round-robin system.

TDM



TDM

Time-Division Multiplexing

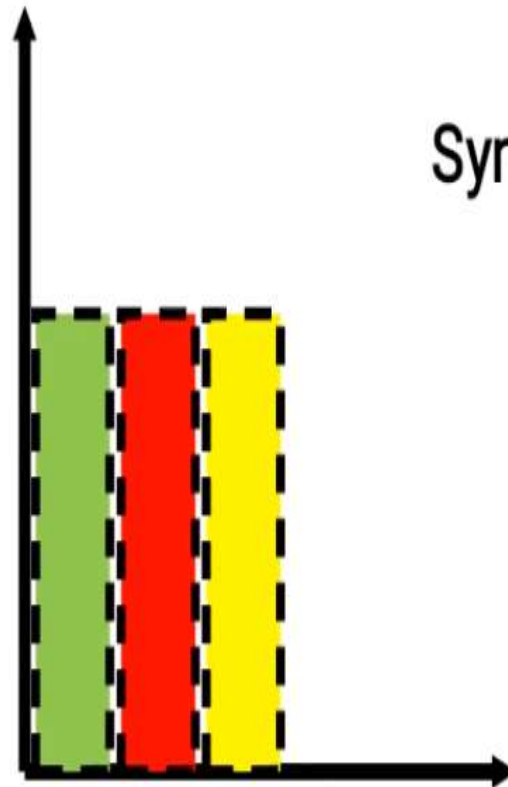


Synchronous TDM



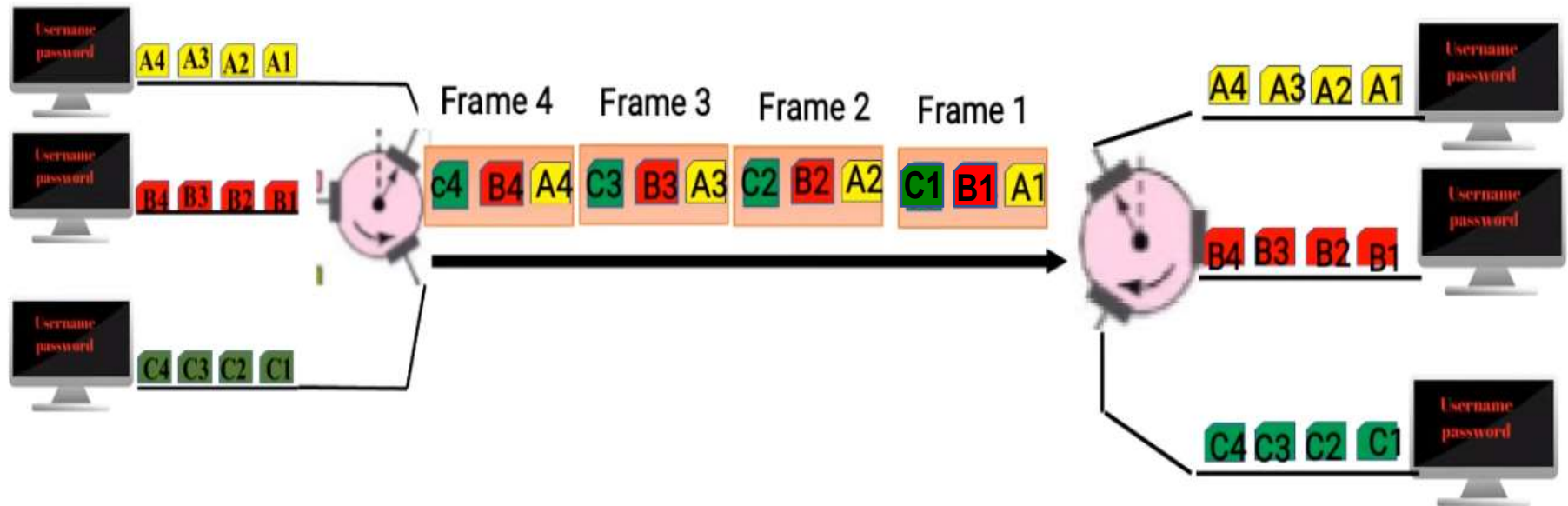
Statistical TDM
/Asynchronous
TDM

Frequency



Time

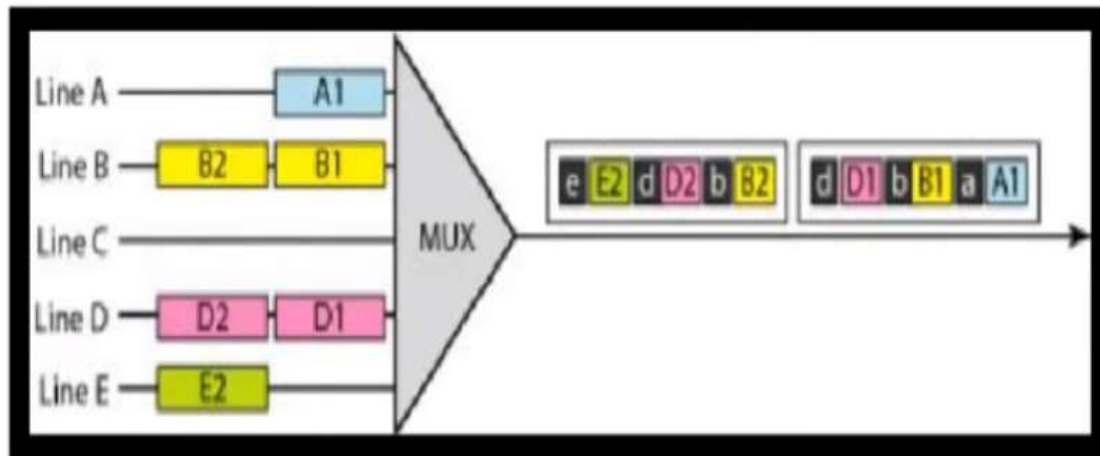
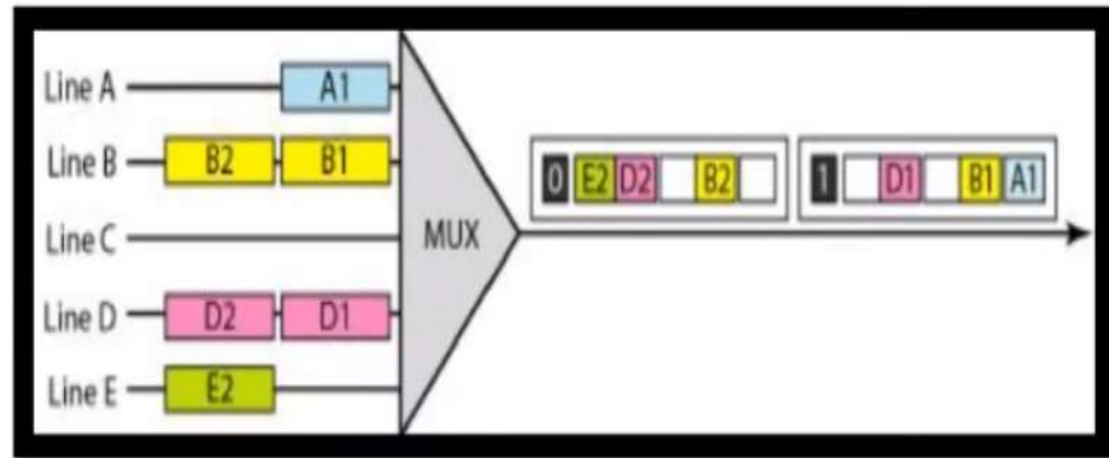
Synchronous TDM



Empty slots If a source does not have data to send, the corresponding slot in the output frame is empty.

Asynchronous TDM

Synchronous TDM



Statistical TDM

TDM

- An example of TDM is digital ground telephone networks (like **ISDN**) .

TDMA

- **Time Division Multiple Access**- TDMA represents Time Division Multiple Access. It is a **cellular mobile technology**.
- It's **an access method for the data link layer**, that uses the concepts of TDM to basically achieve the same goal. In TDM, each sender is provided the whole possession of the whole bandwidth of the medium for a constant duration of time.
- An example of TDMA is **GSM** (actually GSM is a hybrid system containing both TDMA and FDMA) .
- GSM means Global System for Mobile Communication.

FDMA

- **FDMA represents Frequency Division Multiple Access.** It is a **cellular mobile technology**. It's **an access method for the data link layer**, that uses the concepts of FDM to basically achieve the same goal.
- The frequency bands of different stations are separated by small band of unused frequency called as **guard bands** that prevents the interference of stations.
- All the information is combined prior to generating a signal at data link layer, there is **no physical multiplexer at the physical layer**.

FDMA vs. TDMA

FDMA	TDMA
Frequency Division Multiple Access	Time Division Multiple Access
In this bandwidth is shared by different stations all the time	In this whole bandwidth is used by different stations for sometime
Guard bands are necessary between adjacent channels	Guard time is necessary between adjacent time slots
Rate of data is low	Rate of data is medium
Synchronization is not required	Synchronization is required

What is Network Switching?

- **Network switching** is the process of channeling data received from any number of input ports to another designated port that will transmit the data to its desired destination.
- The device through which the input data passes is called a **switch**.
- Data entering a port is referred to as **ingress**, while data leaving the port is referred to as **egress**.
- The switch represents the medium through which the data is routed to its final destination.

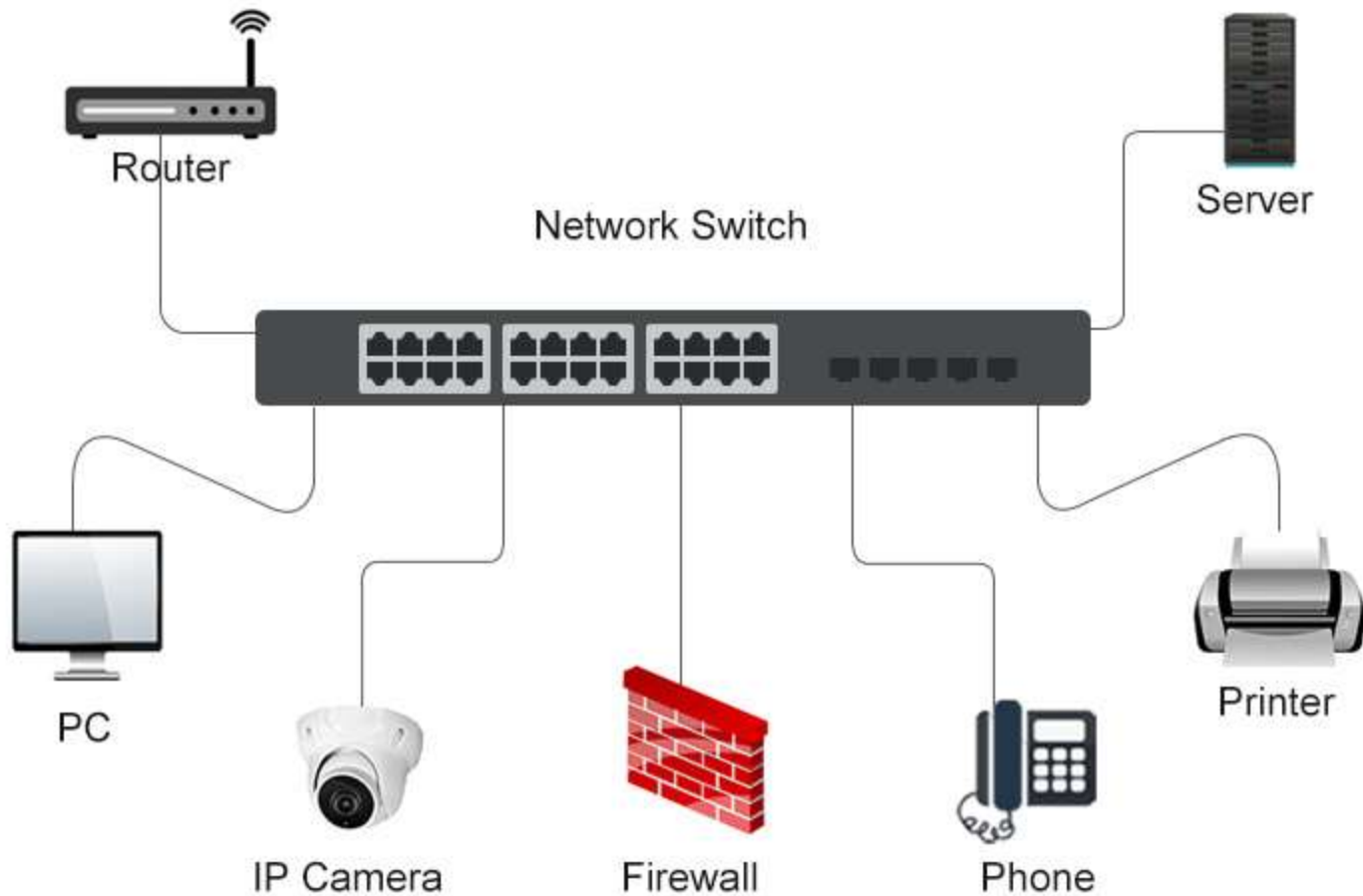
What is a Switch?

- A switch is a **small hardware device** which is used to join multiple computers together with one local area network (LAN).
- Network switches operate at layer 2 (**Data link layer**) in the OSI model.
- Switches are used to forward the data frames based on **MAC addresses**.
- A Switch is used to transfer the data only to the device that has been addressed. Thus, it reduces data traffic.

What is a Switch?



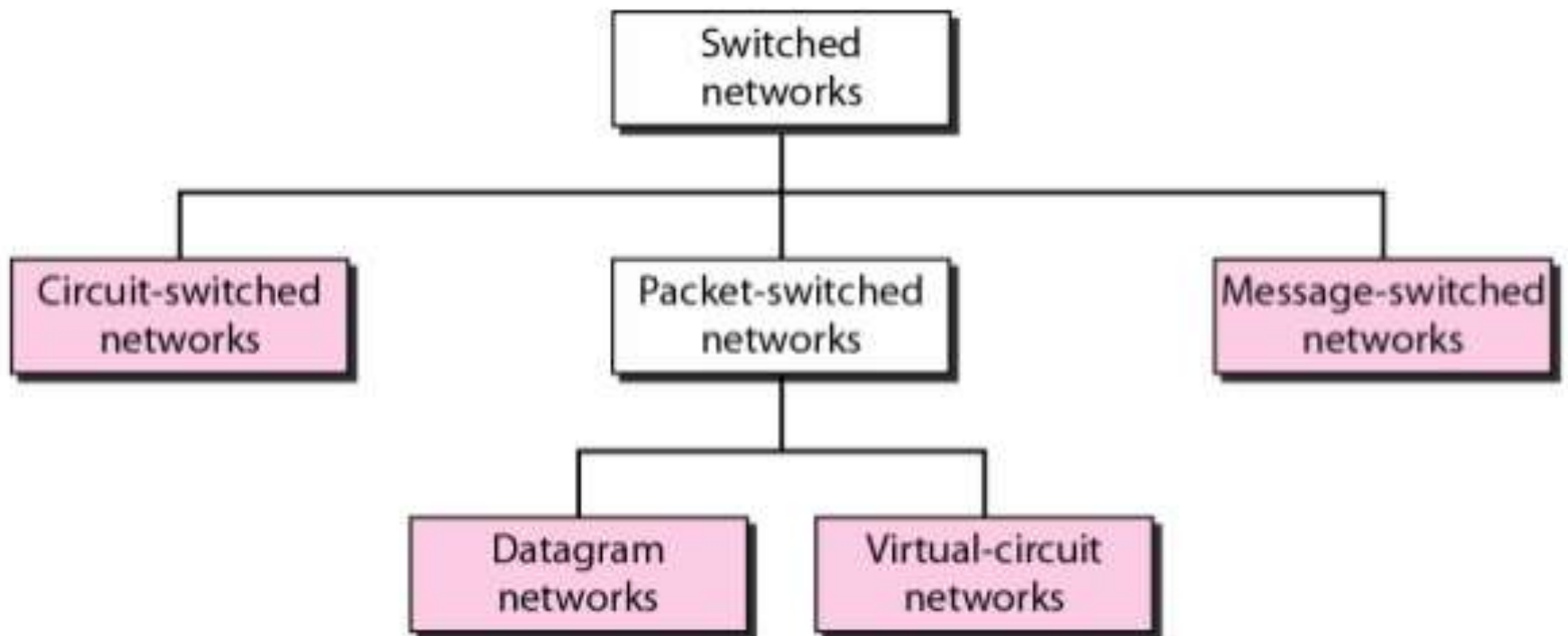
What is a Switch?



Why Switching?

- Switching in computer networks help in deciding **the best route for data transmission** if there are multiple paths in a large network.
- Switching techniques are used for the **effective utilization of the bandwidth** of a network.
- Switching is a full duplex communication channel that's why it **avoids collision** between data packets in a network.

Switched networks



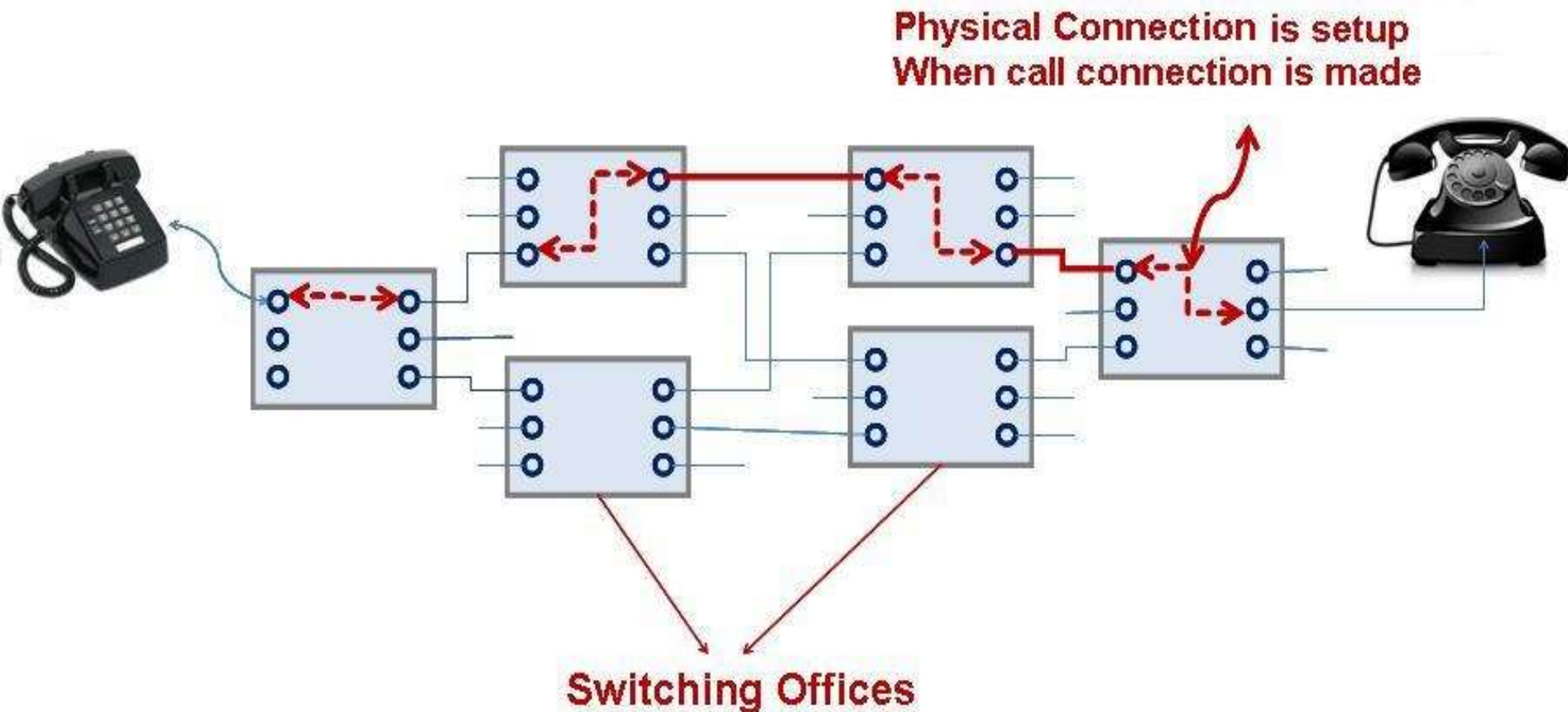
Circuit Switching

- **Circuit switching** is defined as the establishment of a dedicated communication path between the two parties, or nodes, within a physical network.
- This path (circuit) is established and maintained for the duration of the session.
- No matter the length of the communication session, the circuit will remain and the data paths maintained. The circuit is only terminated when the session ends.
- This type of network switching is well suited for **voice communications**.

Circuit Switching

- Example: Telephone network.
- 3 phases in Circuit Switching
 - **Connection establishment**
 - **Data Transfer**
 - **Connection Disconnection**

Circuit Switching



Advantages of Circuit Switching

- Once the circuit is established there is no delay in data transfer. So it is **useful in real-time data transfer like voice calls.**
- The communication channel is dedicated which ensures a steady rate of data transfer.
- There is no need of packet sorting in the receiver's side because all the nodes follow the same path and in the same order.

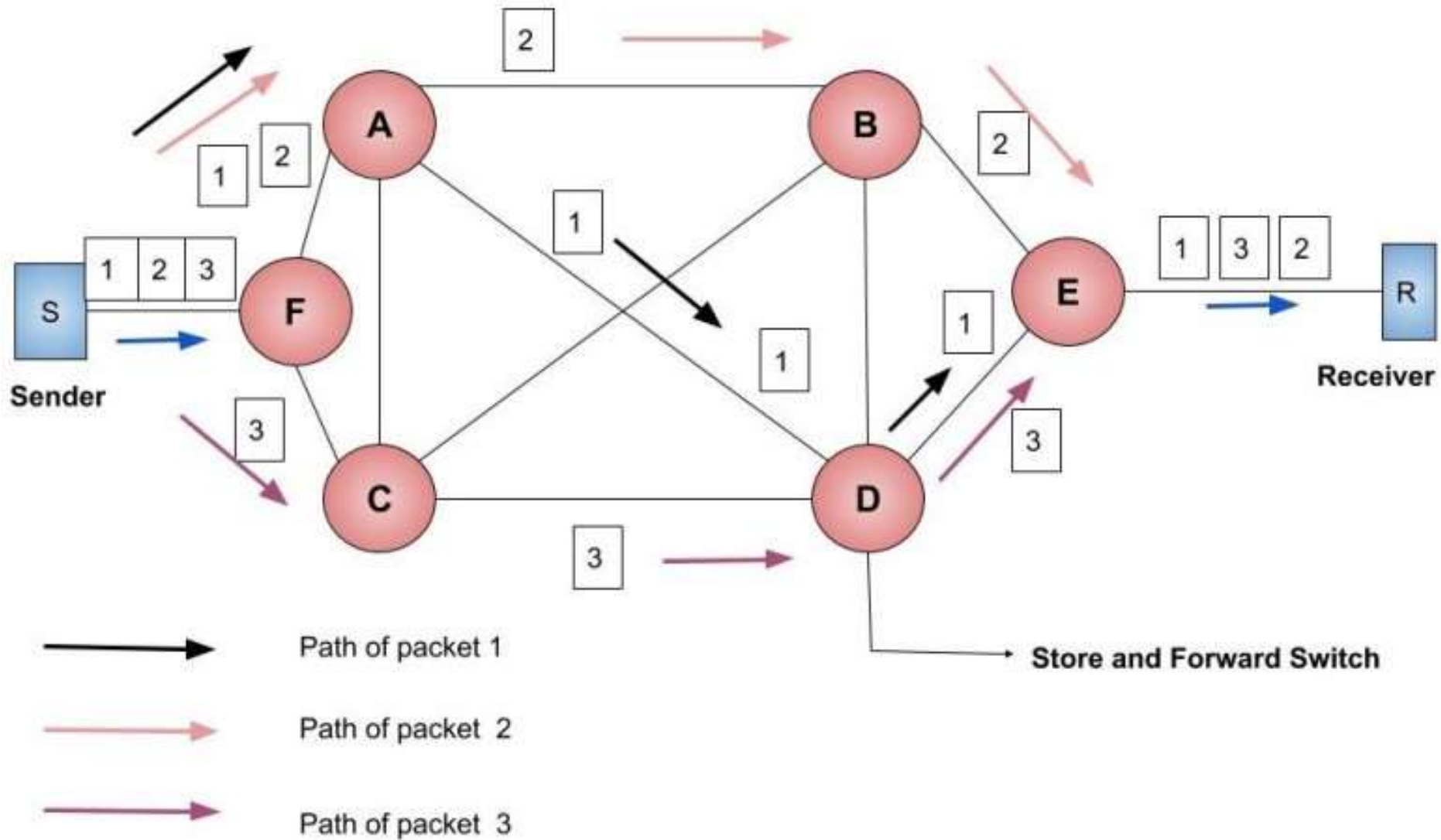
Disadvantages of Circuit Switching

- As the **channel is reserved** it cannot be used to transfer any other data even if the connected systems are not using the channel. Hence circuit switching is **insufficient in terms of resource utilization**.
- It is **expensive** as compared to other techniques because of dedicated path requirements.
- The time required to establish the connection between the source and destination is high. This means that there is no communication until the connection is established.
- It is not suitable for data transmission.

Packet Switching

- Packet switching is a message switching technique in which the data is divided into **packets**. These packets contain a header that contains the information of the source and destination. Each packet then independently find the route with the help of this information.
- The packet belonging to the same file may **take different paths depending upon the availability of the path**. At the destination, these packets are **re-assembled**. It is the responsibility of the receiver to re-arrange the received packet to the original data.
- The biggest packet-switched network is the **internet**.

Datagram Packet Switching



Advantages of Datagram Packet Switching

- It is more efficient for data transmission as it doesn't require any dedicated paths.
- If the link is busy or not available, the packets can be re-routed. This ensures a reliable connection.
- The same channel can be used by many users simultaneously.
- With improved protocols, packet switching is used for applications like **Skype, WhatsApp**, etc.

Disadvantages of Datagram Packet Switching

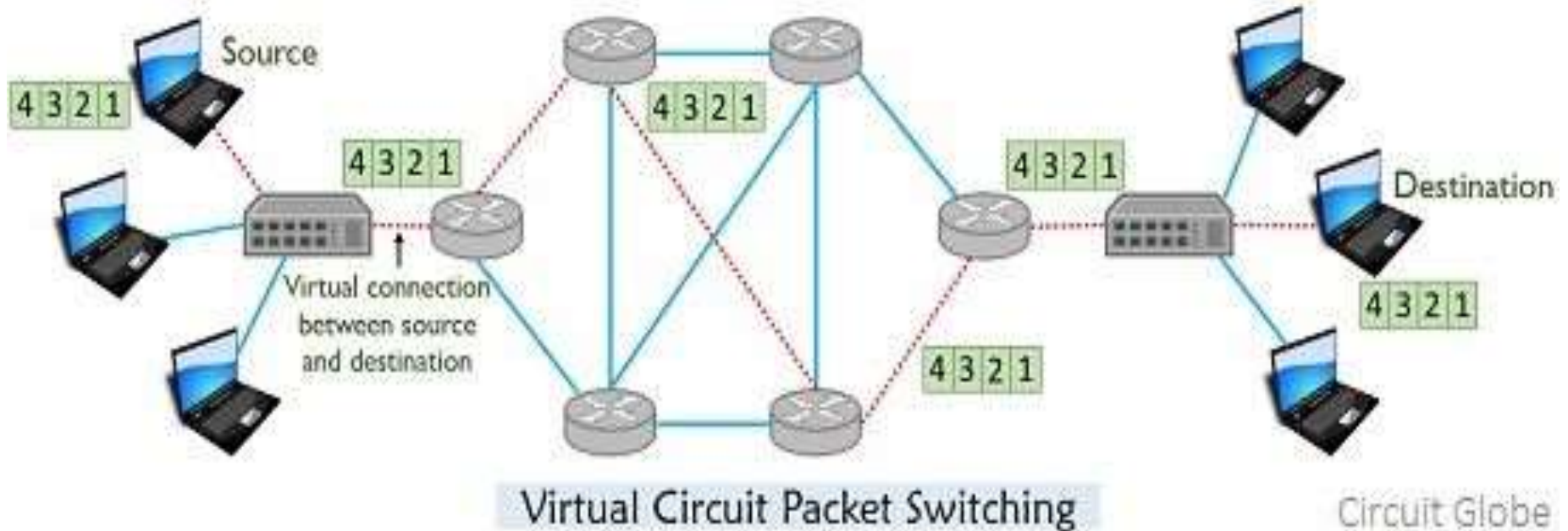
- They cannot be used for applications that cannot afford delays like high-quality voice calls.
- Protocols used in packet-switching are complex and require higher implementation costs.
- If the network is overloaded then the packets may be lost or delayed. This may lead to loss of critical information.
- Sorting of received packets is required at the receiver's side.

Virtual Circuit Switching

Virtual Circuit switching is the type of packet switching which is connection-oriented network. In virtual circuit resource are reserve for the time interval of data transmission between two nodes.

- A preplanned route is established before the messages are sent.
- This network is a highly reliable medium of transfer. Virtual circuits are costly to implement.

Virtual Circuit Packet Switching



Advantages of Virtual Circuit Packet Switching

- Packets are delivered to the receiver in the same order sent by the sender.
- Virtual circuit is a reliable network circuit.
- There is no need for overhead in each packet.
- Single global packet overhead is used in virtual circuit.

Disadvantages of Virtual Circuit Switching

- Virtual circuit is costly to implement as switching equipments need to be more powerful.
- It provides only connection-oriented service.
- Always a new connection set up is required for transmission.

Circuit Switching vs. Packet Switching

Circuit Switching	Packet Switching
In-circuit switching there are 3 phases: i) Connection Establishment. ii) Data Transfer. iii) Connection Released.	In Packet switching directly data transfer takes place.
In-circuit switching, each data unit knows the entire path address which is provided by the source.	In Packet switching, each data unit just knows the final destination address intermediate path is decided by the routers.

Circuit Switching vs. Packet Switching

Circuit Switching	Packet Switching
Resource reservation is the feature of circuit switching because the path is fixed for data transmission.	There is no resource reservation because bandwidth is shared among users.
More Wastage of resources	Less Wastage of resources
More reliable	Less reliable

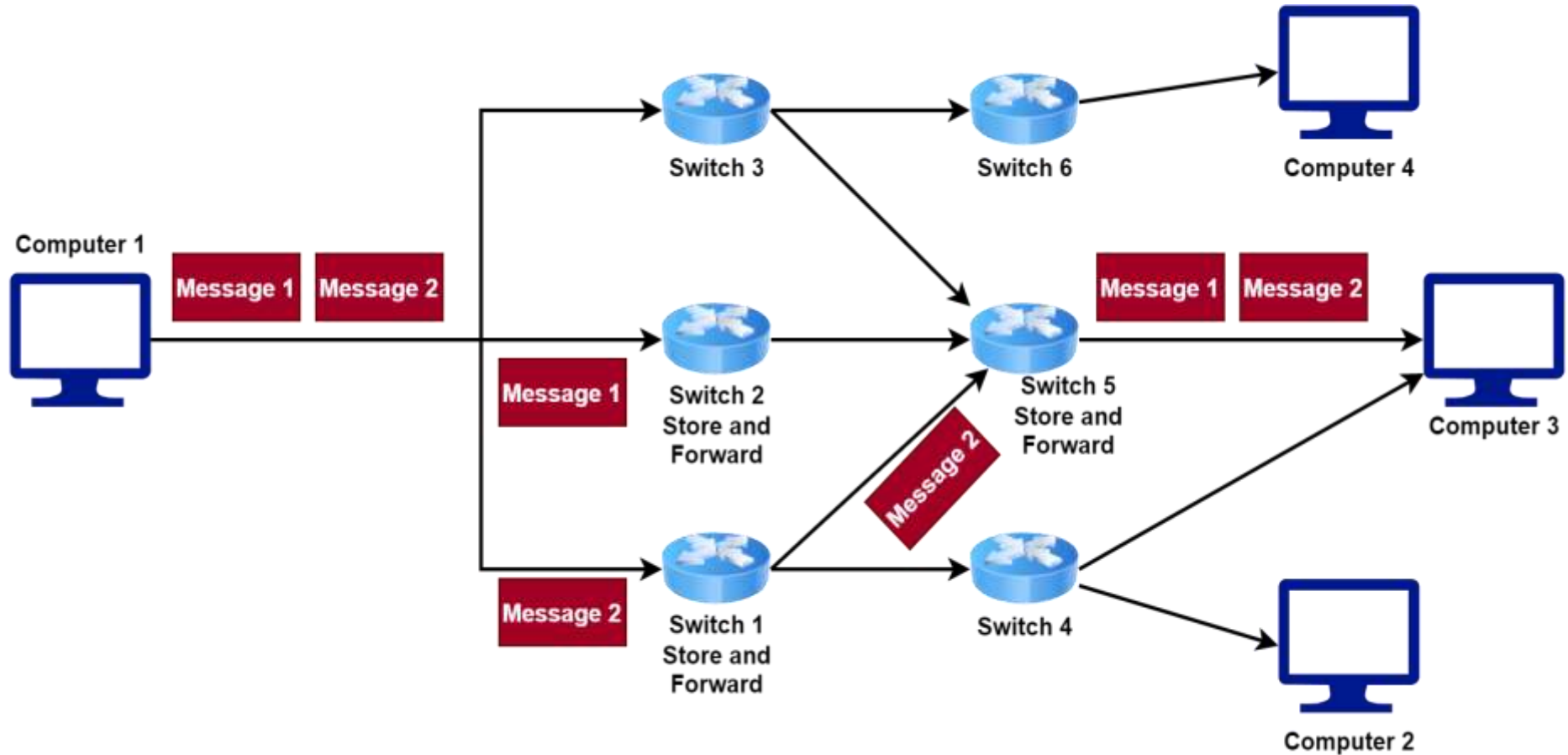
Circuit Switching vs. Packet Switching

Circuit Switching	Packet Switching
In-circuit switching each packet follows the same route.	In packet switching packets can follow any route.
The circuit switching network is implemented at the physical layer.	Packet switching is implemented at the datalink layer and network layer
Circuit switching requires simple protocols for delivery.	Packet switching requires complex protocols for delivery.

Message Switching

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- There is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.

Message Switching



Advantages Message Switching

- Traffic congestion can be reduced because the message is temporarily stored in the nodes.
- Message priority can be used to manage the network.
- The size of the message which is sent over the network can be varied. Therefore, it supports the data of unlimited size.

Disadvantages Message Switching

- The message switches must be equipped with sufficient storage to enable them to store the messages until the message is forwarded.
- The Long delay can occur due to the storing and forwarding facility provided by the message switching technique.

Data Transmission Media

- In data communication terminology, a transmission medium is a **physical path between the transmitter and the receiver** i.e it is the channel through which data is sent from one place to another.
- The main functionality of the transmission media is to carry the information in the form of bits through **LAN**(Local Area Network).
- In **OSI**(Open System Interconnection) phase, **transmission media supports the Layer 1**. Therefore, it is considered to be as a Layer 1(Physical Layer) component.

Data Transmission Media

- The electrical signals can be sent through the copper wire, fibre optics, atmosphere, water, and vacuum.
- The characteristics and quality of data transmission are determined by the characteristics of medium and signal.
- Transmission media are of two types - **wired media and wireless media**. In wired media, medium characteristics are more important whereas, in wireless media, signal characteristics are more important.
- Different transmission media have different properties** such as bandwidth, delay, cost and ease of installation and maintenance.

Factors important for Data Transmission

- **Bandwidth:** The greater the bandwidth of a medium, the higher the data transmission rate of a signal.
- **Transmission impairment:** When the received signal is not identical to the transmitted one due to the transmission impairment(noise or attenuation), the quality of the signals will get destroyed.
- **Interference:** An interference is defined as the process of **disrupting a signal when it travels over a communication medium** on the addition of some unwanted signal.

Data Transmission Media

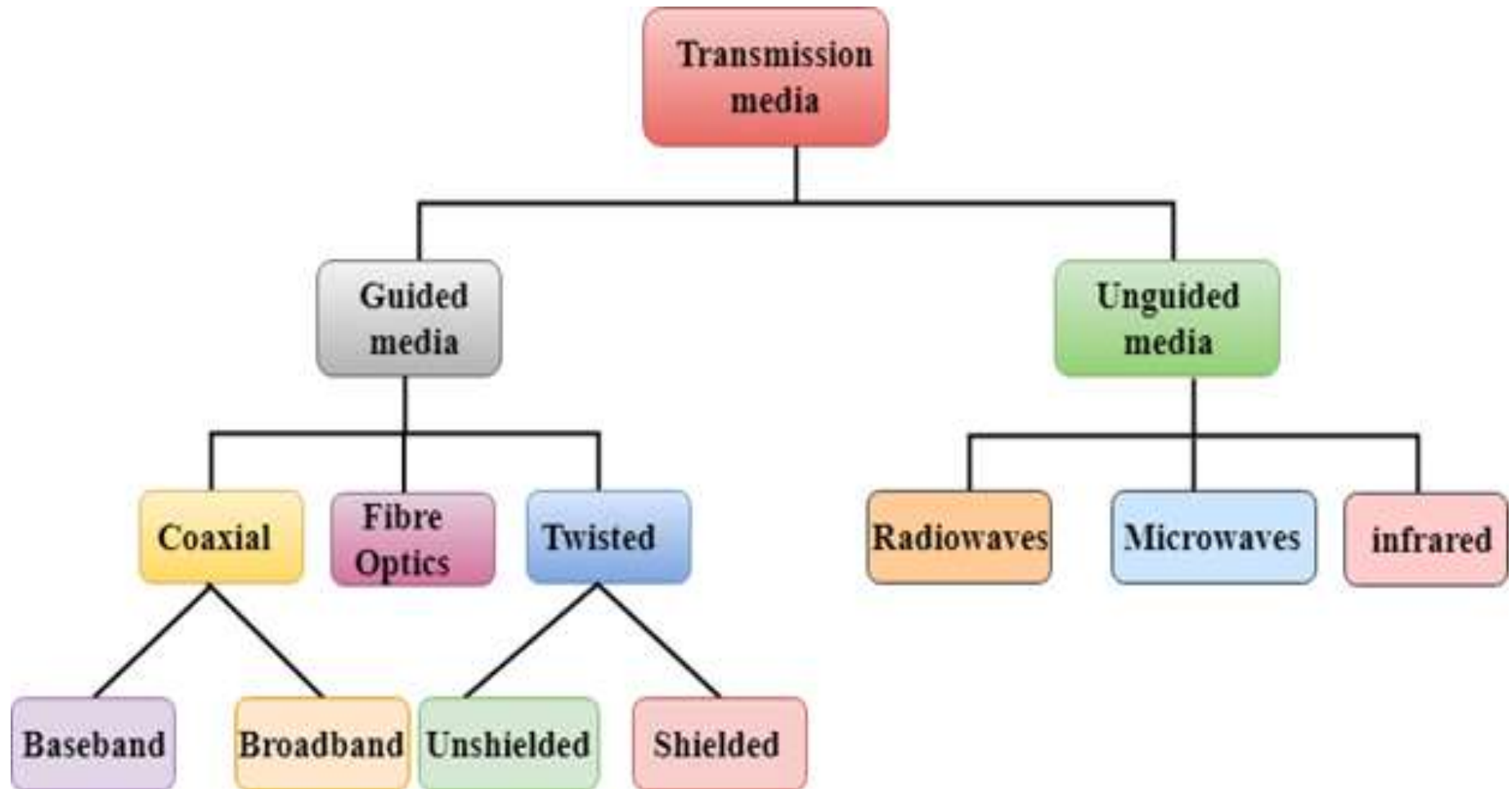
- Physical conductors have certain limitations so Guided Media will also have its limitations such as:

- Length of the conductor

- Installation cost

- Maintenance

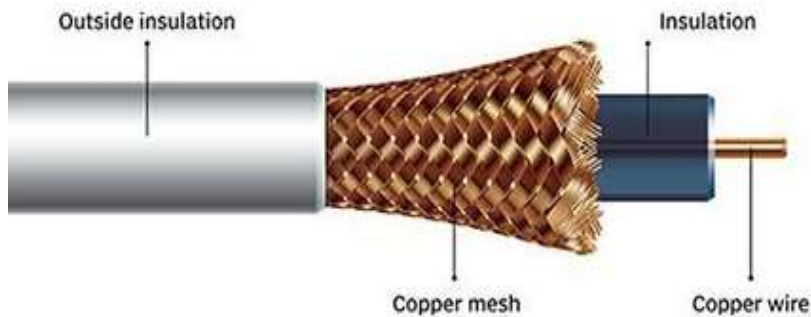
Classification of Data Transmission



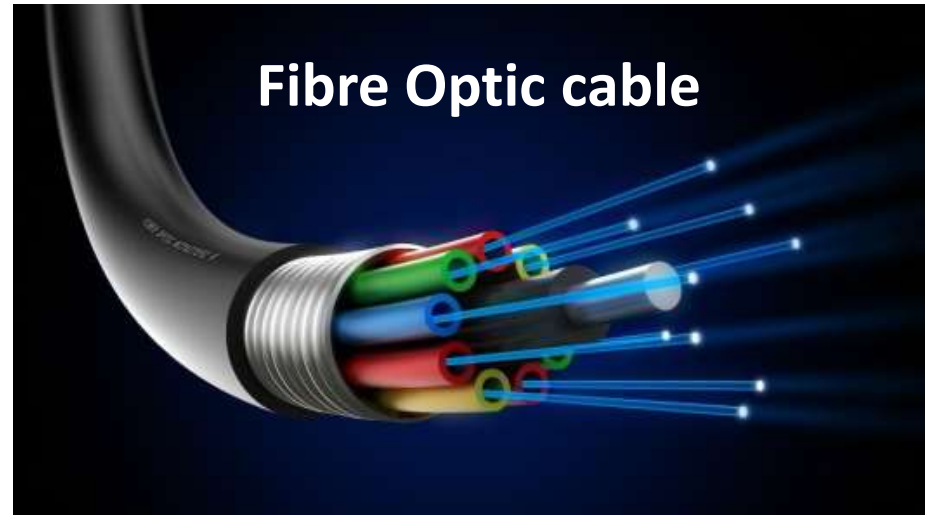
Guided Transmission Media

- It is defined as the physical medium through which the signals are transmitted. It is also known as Bounded media.

Coaxial cable



Fibre Optic cable



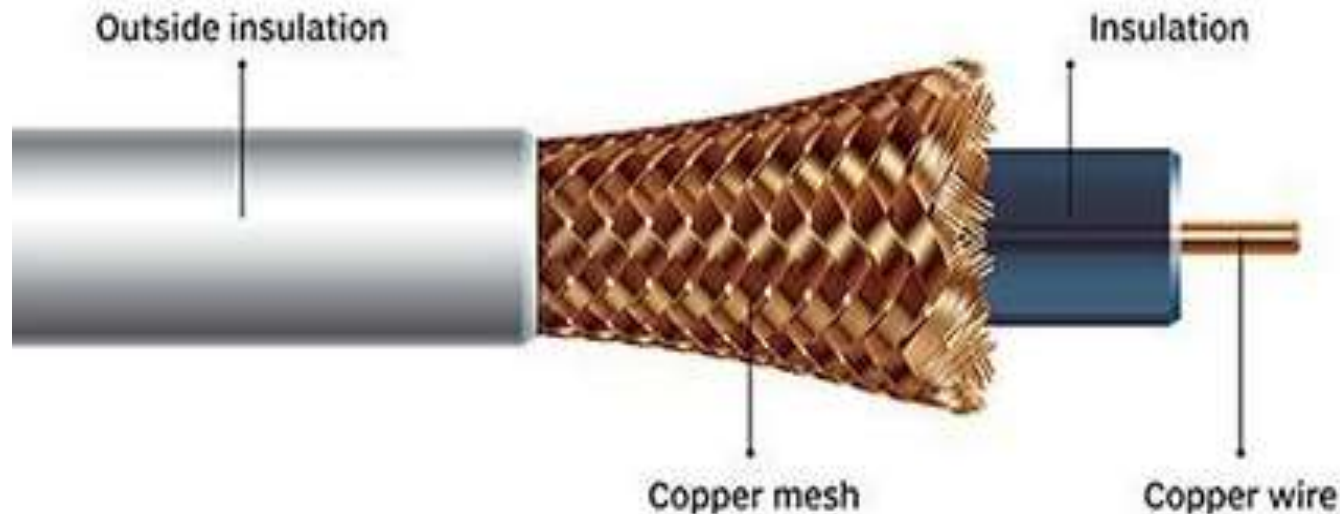
Twisted Pair cable



Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.

Coaxial cable



Coaxial Cable

- It has a higher frequency as compared to Twisted pair cable.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the **EMI**(Electromagnetic interference).

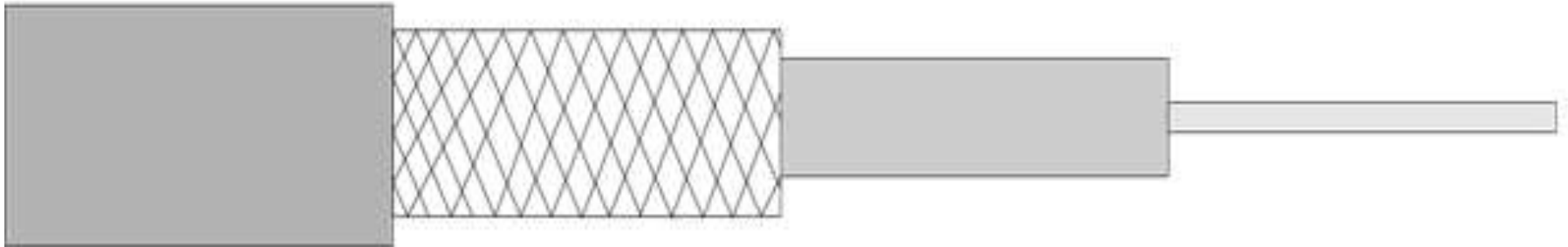
Diagram of Coaxial Cable

Jacket

Shield

Insulator

Centre Conductor



Coaxial cable is of two types:

Baseband transmission: It is defined as the process of transmitting a single signal at high speed.

Broadband transmission: It is defined as the process of transmitting multiple signals simultaneously.

Advantages of Coaxial Cable

- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages of Coaxial Cable

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

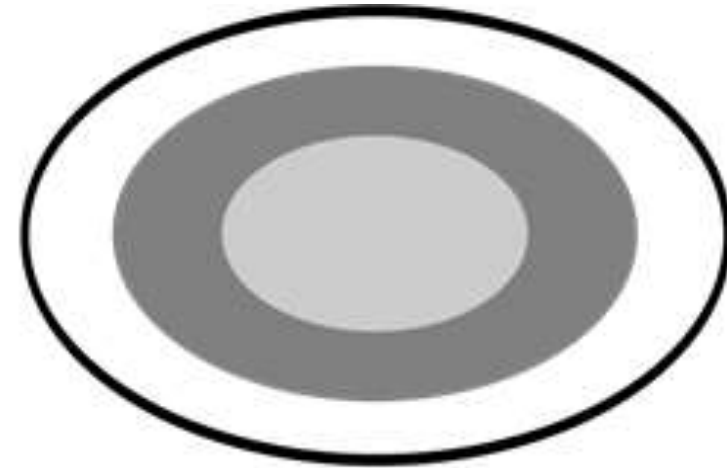
Fibre Optic Cable

- A Fibre optic cable is made up of high quality thin glass or plastic and is used to transfer digital data signals in the form of light upto distance of thousand of miles.
- The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring so noise and distortion is very less.
- Fibre optics provide faster data transmission than copper wires.

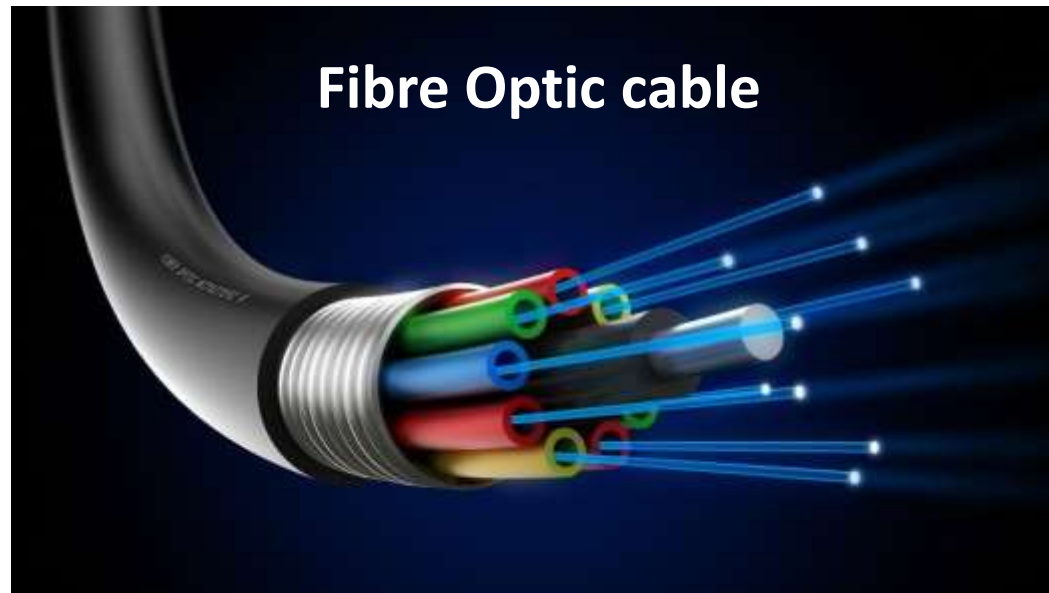
Diagram of Fibre Optic Cable



Side View



End View



Basic Elements of Fibre Optic Cable

- Core:** The optical fibre consists of a narrow strand of glass or plastic known as a core. **A core is a light transmission area of the fibre.** The more the area of the core, the more light will be transmitted into the fibre.
- Cladding:** The core is surrounded by the layer of glass called cladding. **It reflects light inwards to avoid loss of signal and allow the light to pass through bends in the cable. No light escapes the glass core.**
- Jacket:** The protective coating consisting of plastic is known as a jacket. **The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.**

Advantages of Fibre Optic Cable

- **Greater Bandwidth:** The fibre optic cable provides more bandwidth as compared copper. Therefore, the fibre optic carries more data as compared to copper cable.
- **Faster speed:** Fibre optic cable carries the data in the form of light. This allows the fibre optic cable to carry the signals at a higher speed.
- **Longer distances:** The fibre optic cable carries the data at a longer distance as compared to copper cable.

Advantages of Fibre Optic Cable

- **Better reliability:** The fibre optic cable is more reliable than the copper cable as it is immune to any temperature changes while it can cause obstruct in the connectivity of copper cable.
- **Thinner and Sturdier:** Fibre optic cable is thinner and lighter in weight so it can withstand more pull pressure than copper cable.

Twisted Pair Cable

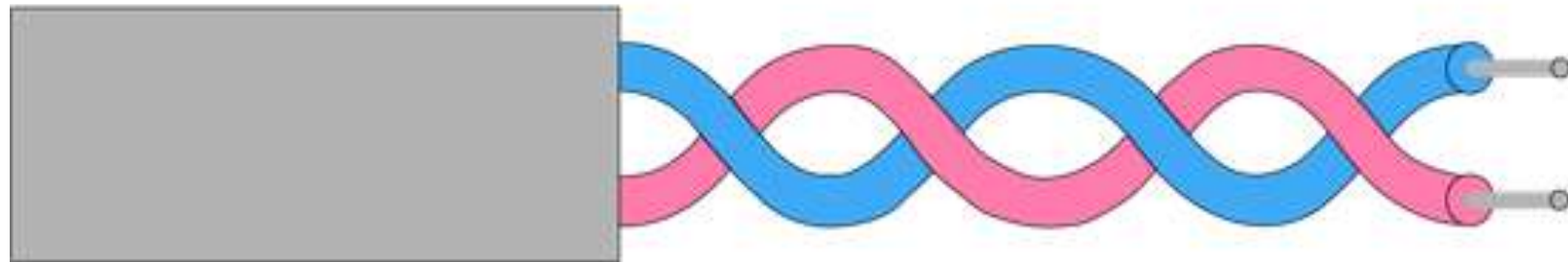
- Twisted pair is a physical media made up of a pair of cables twisted with each other. A twisted pair cable is **cheap** as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable. The **frequency range for twisted pair cable is from 0 to 3.5KHz.**
- A twisted pair consists of two insulated copper wires **arranged in a regular spiral pattern.**
- **The degree of reduction in noise interference is determined by the number of turns per foot.** Increasing the number of turns per foot decreases noise interference.

Diagram of Twisted Pair Cable

Jacket

Twisted Pair

Bare Wire



Twisted Pair

Unshielded Twisted Pair

Shielded Twisted Pair

Unshielded vs Shielded



VS



Unsheilded Twisted Pair Cable

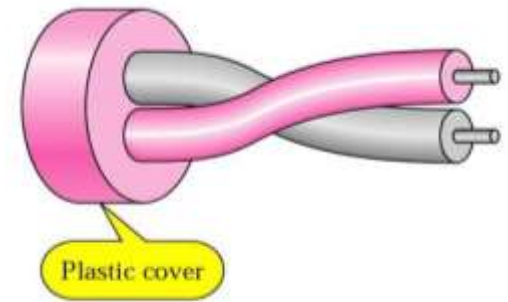
This type of cable has the ability to block interference and does not depend on a physical shield for this purpose. It is used for telephonic applications.

Advantages :

- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.
- It is cheap.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.



a. UTP

Shielded Twisted Pair Cable

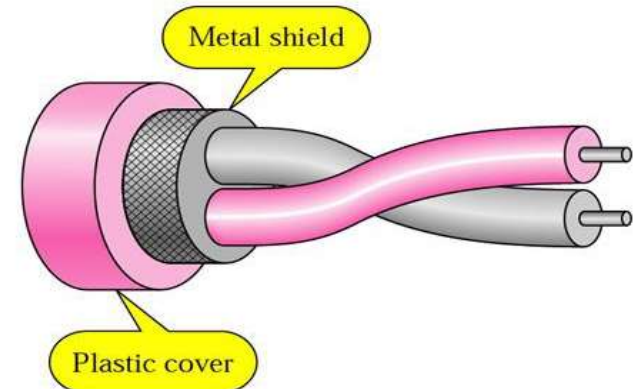
This type of cable consists of a special jacket to block external interference. It is used in fast-data-rate Ethernet and in voice and data channels of telephone lines.

Advantages:

- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster

Disadvantages :

- It has a higher attenuation rate.
- It is more expensive as compared to UTP and coaxial cable.



b. STP

Unguided Media

An unguided transmission transmits the **electromagnetic waves** without using any physical medium. Therefore it is also known as **wireless transmission**. (**Air is the medium**)



Television

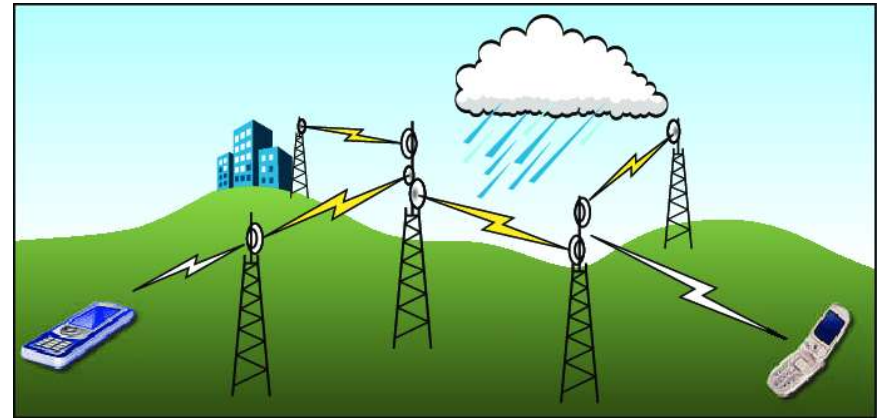


Infrared Radiations

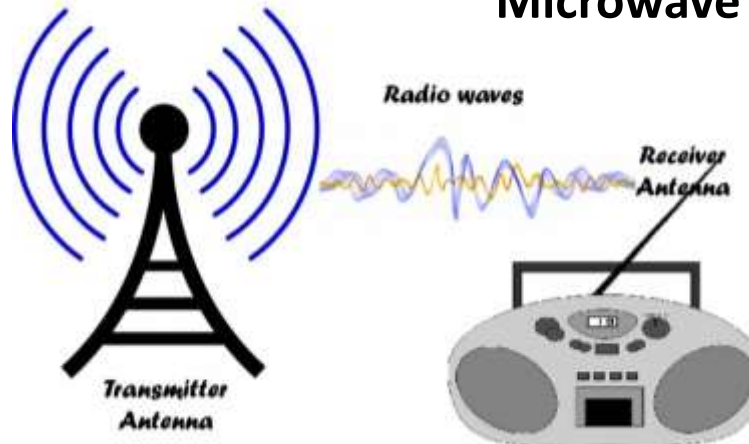


Remote

Infrared Transmission

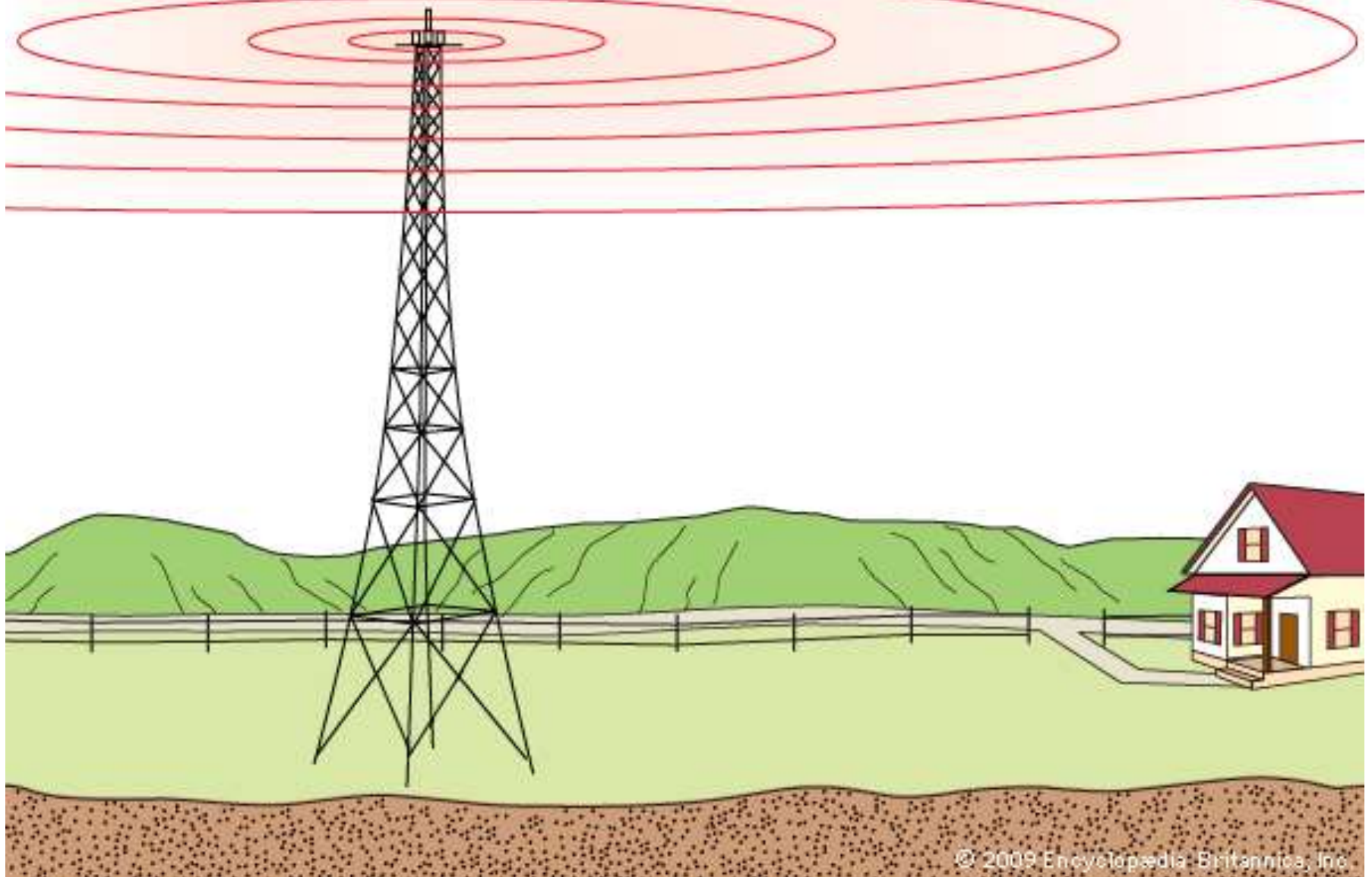


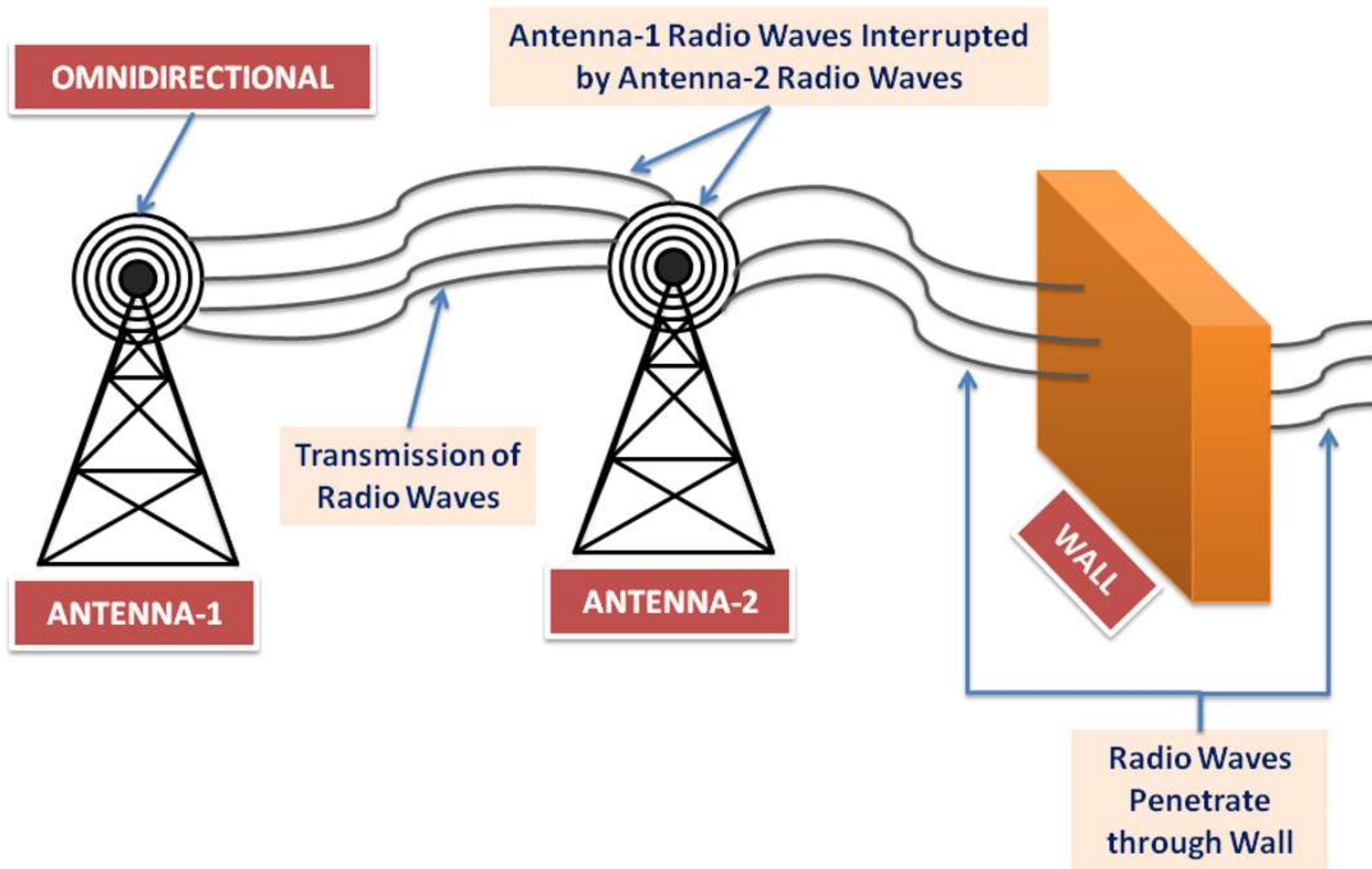
Microwave Transmission



Radiowave Transmission

Radio Waves





Radio Waves

- Radio waves are the electromagnetic waves that are transmitted in all the directions of free space.
- **Radio waves are omnidirectional**, i.e., the signals are propagated in all the directions.
- The range in frequencies of radio waves is from 3Khz to 1 Ghz.
- In the case of radio waves, the sending and receiving antenna are not aligned, i.e., **the wave sent by the sending antenna can be received by any receiving antenna.**
- An example of the radio wave is **FM radio.**

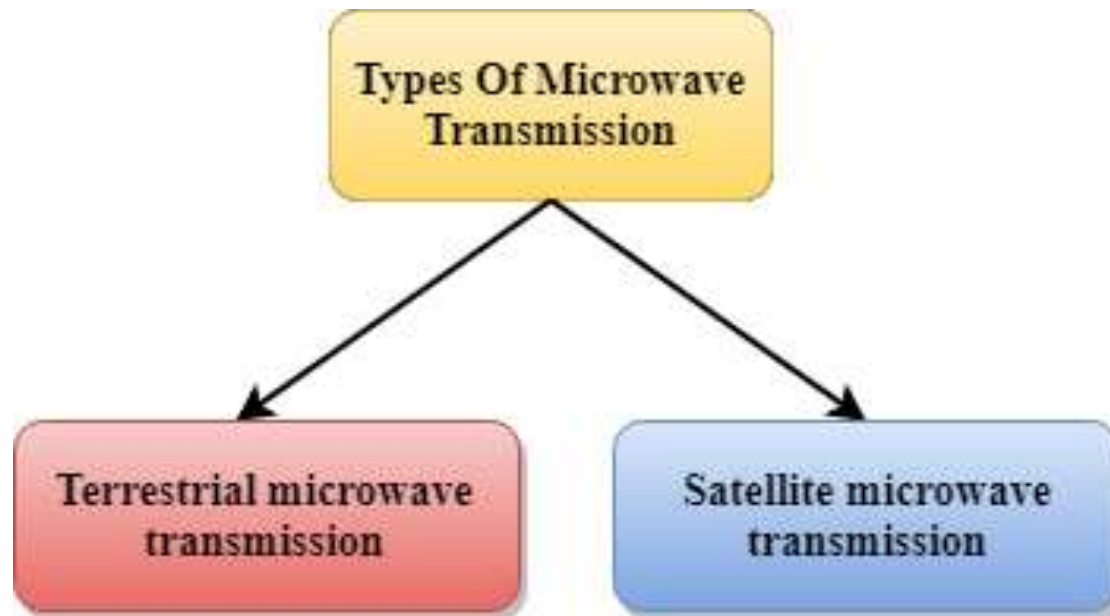
Application of Radio Waves

- A Radio wave is useful for multicasting when there is one sender and many receivers.
- An FM radio and televisions are examples of a radio wave.

Advantages of Radio Waves

- Radio transmission is mainly used for wide area networks.
- Radio waves cover a large area, and they can penetrate the walls.
- Radio transmission provides a higher transmission rate.

Microwaves

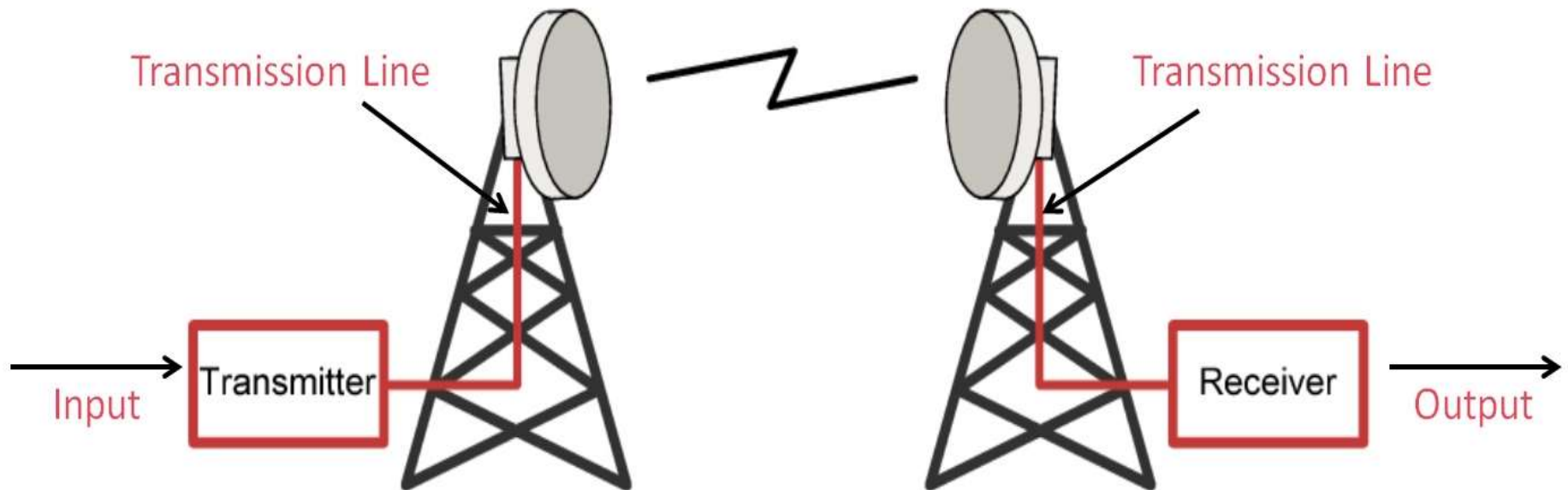


Terrestrial Microwave Transmission

- Terrestrial Microwave transmission is a technology that transmits the focused beam of a radio signal from one ground-based microwave transmission antenna to another.
- Microwaves are the electromagnetic waves having the frequency in the range from 1GHz to 300 GHz.
- Microwaves are unidirectional as the sending and receiving antenna is to be aligned, i.e., the waves sent by the sending antenna are narrowly focused.

Terrestrial Microwave Transmission

- In this case, antennas are mounted on the towers to send a beam to another antenna which is km away.
- It works on the line of sight transmission, i.e., the antennas mounted on the towers are in direct sight of each other.



Advantages of Microwave Transmission

- Microwave transmission is cheaper than using cables.
- It is free from land acquisition as it does not require any land for the installation of cables.
- Microwave transmission provides an easy communication in terrains as the installation of cable in terrain is quite a difficult task.
- Communication over oceans can be achieved by using microwave transmission.

Disadvantages of Microwave Transmission

- Eavesdropping:** An eavesdropping creates insecure communication. Any malicious user can catch the signal in the air by using its own antenna.
- Out of phase signal:** A signal can be moved out of phase by using microwave transmission.
- Susceptible to weather condition:** A microwave transmission is susceptible to weather condition. This means that any environmental change such as rain, wind can distort the signal.
- Bandwidth limited:** Allocation of bandwidth is limited in the case of microwave transmission.

Satellite Microwave Communication

- A satellite is a physical object that revolves around the earth at a known height.
- Satellite communication is more reliable nowadays as it offers more flexibility than cable and fibre optic systems.
- We can communicate with any point on the globe by using satellite communication.

Satellite Microwave Communication

How Does Satellite work?

The satellite accepts the signal that is transmitted from the earth station, and it amplifies the signal. The amplified signal is retransmitted to another earth station.



Advantages of Satellite Microwave Communication

- The coverage area of a satellite microwave is more than the terrestrial microwave. It is easy to install.
- The transmission cost of the satellite is independent of the distance from the centre of the coverage area.
- Satellite communication is used in mobile and wireless communication applications.
- It is used in a wide variety of applications such as weather forecasting, radio/TV signal broadcasting, mobile communication, etc.

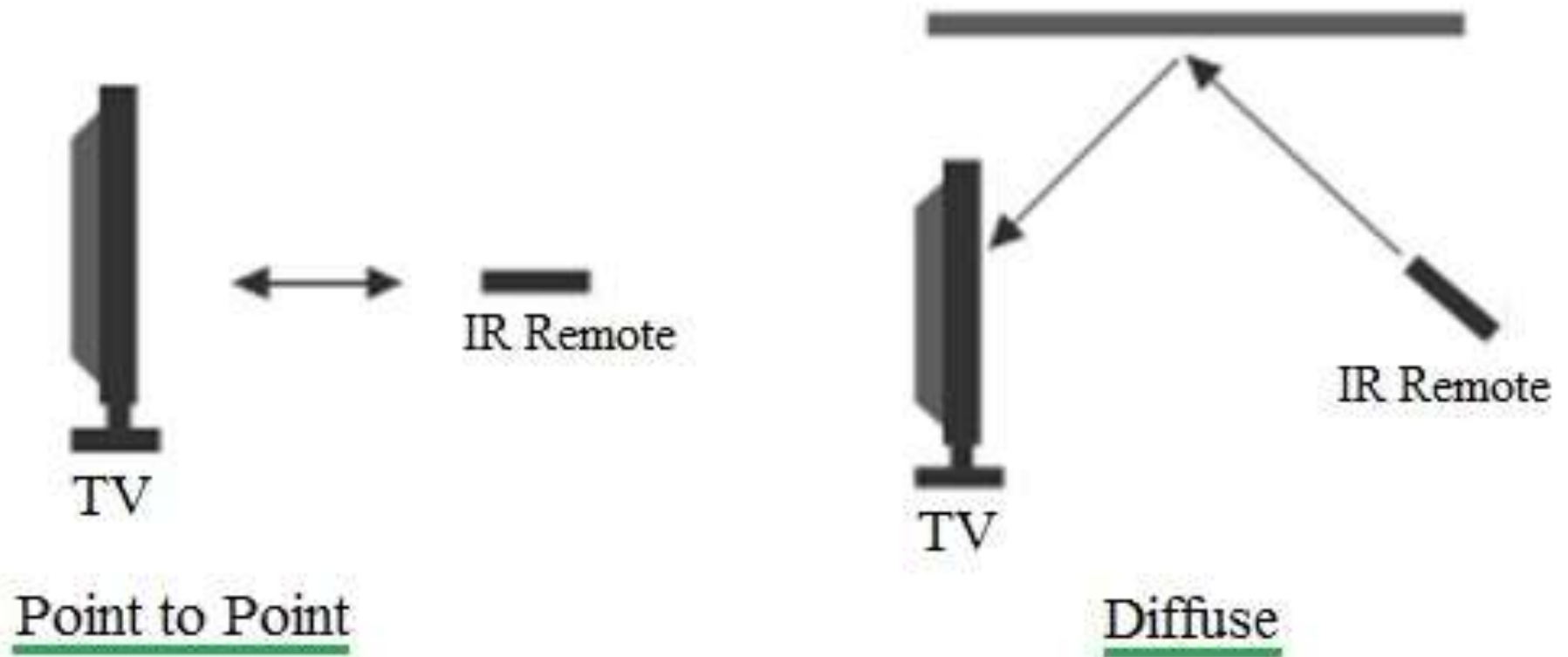
Disadvantages of Satellite Microwave Communication

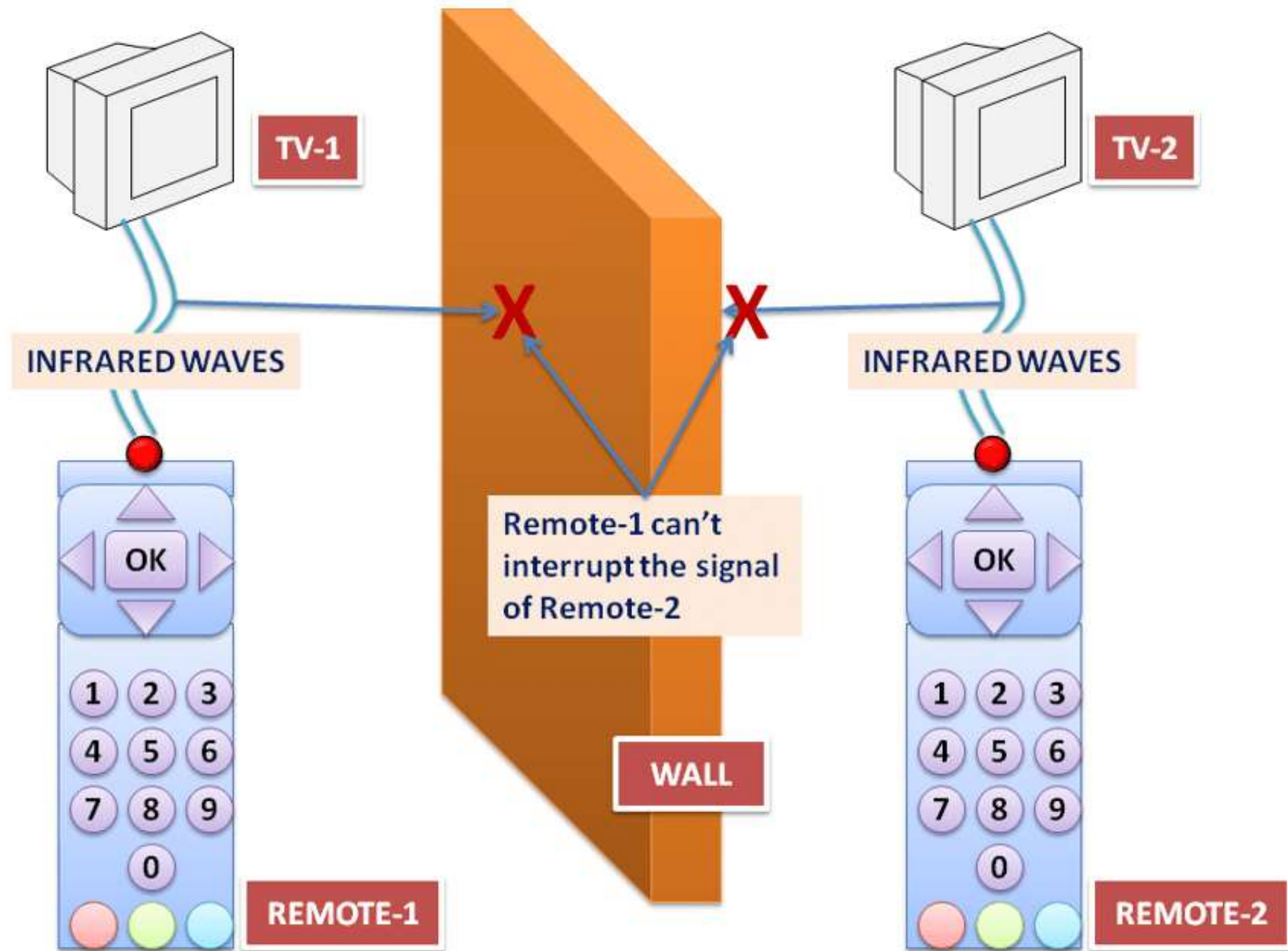
- Satellite designing and development requires more time and higher cost.
- The Satellite needs to be monitored and controlled on regular periods so that it remains in orbit.
- The life of the satellite is about 12-15 years. Due to this reason, another launch of the satellite has to be planned before it becomes non-functional.

Infrared

- An infrared transmission is a wireless technology used for communication over short ranges.
- The frequency of the infrared is in the range from 300 GHz to 400 THz.
- It is used for short-range communication such as data transfer between two cell phones, TV remote operation, data transfer between a computer and cell phone resides in the same closed area.

Infrared





Characteristics of Infrared

- **It supports high bandwidth**, and hence the data rate will be very high.
- **Infrared waves cannot penetrate the walls**. Therefore, the infrared communication in one room cannot be interrupted by the nearby rooms.
- **An infrared communication provides better security** with minimum interference.
- **Infrared communication is unreliable outside the building** because the sun rays will interfere with the infrared waves.